SUPPLIER CERTIFICATE OF ANALYSIS (COA) & SPC CPK REPORT

1. INCOMING LOT IDENTIFICATION & MATERIAL TRACEABILITY

- Lot / Batch Number: BATCH-XYZ-AUST-8841 | Supplier: AseptiPack Foil & Closures (SUP-XYZ-T1-ASEPTIPACK).
- Target Finished Good: SKU-XYZ-204 (Aseptic HydraPure Electrolyte Beverage 24-Pack) at PLANT-XYZ-AUSTIN.
- Inspection Method: Document AI OCR Extraction + Inline GigE 120-FPS Optical Vision Verification.

2. STATISTICAL PROCESS CONTROL (SPC) & CPK DRIFT FINDINGS

- Induction Foil Laminate Thickness: Spec 38.0 ± 2.0 microns | Measured Mean: 36.4 microns | Cpk = 1.18 (DRIFT ALERT).
- Tensile Seal Burst Pressure: Spec >= 4.2 Bar | Measured: 4.45 Bar (PASS).
- Action Taken: QualiShield placed Lot 8841 on Conditional Hold in SAP QM and issued SCAR-2026-119 + $14,200 COPQ Debit.

STRUCTURED ENTERPRISE LEDGER & VERIFICATION TABLE

Parameter	Engineering Spec	Measured Value	SPC Cpk	Lot Disposition
Foil Thickness	38.0 +/- 2.0 um	36.4 um	1.18 (Drift)	CONDITIONAL_HOLD
Burst Pressure	>= 4.20 Bar	4.45 Bar	1.64 (Nominal)	PASSED
Polymer Coating	12.0 +/- 0.8 g/m2	11.9 g/m2	1.71 (Nominal)	PASSED
Heat Seal Tear	0.0% Micro-Leaks	1 Unit Diverted	98.5% Vision Conf	GATE_DIVERTED

DIGITAL CRYPTOGRAPHIC AUDIT SEAL & ENTERPRISE SIGN-OFF

Organization: XYZ Global Industries Corp. (XYZ-CORP-01) | ERP Sync: SAP S/4HANA + Cloud Spanner Twin

HMAC-SHA256 Verification Hash: 9f84c2a10e7b4d33a81e6c992b10d4f7e3a2118c6d004f9a | Status: APPROVED & VERIFIED

Authorized By: Elena Rostova (CSCO) & Vertex AI Agent Engine Autonomous Governance Enclave

4. CROSS-PILLAR AGENT HANDOFF & GOLDEN THREAD TRACEABILITY

- This document is automatically indexed in AlloyDB pgvector and linked to BigQuery dataset `supply_chain_twin`.
- Shared Enterprise SKUs referenced: SKU-XYZ-101, SKU-XYZ-204, SKU-XYZ-309, SKU-XYZ-412, SKU-XYZ-518, SKU-XYZ-625.
- Shared Facilities referenced: PLANT-XYZ-STUTTGART, PLANT-XYZ-AUSTIN, PLANT-XYZ-BENGALURU, RDC-XYZ-CHICAGO, FDC-XYZ-MUMBAI.

5. GOVERNANCE, MODEL ARMOR & DETERMINISTIC QUALITY GATES

- All agent write-backs require pre-execution `enforce_tool_contract()` validation and a unique `idempotency_key`.
- PII, vendor bank coordinates, and GSTIN/tax identifiers are tokenized via Google Cloud DLP prior to LLM context injection.

STRUCTURED ENTERPRISE LEDGER & VERIFICATION TABLE

Golden Thread ID	SKU ID	Purchase Order	Production Batch	Sales Order / Shipment
GT-XYZ-2026-001	SKU-XYZ-101	PO-XYZ-880142	BATCH-XYZ-STUT-904	SO-XYZ-20261022-801 / SHP-XYZ-90
GT-XYZ-2026-002	SKU-XYZ-204	PO-XYZ-880219	BATCH-XYZ-AUST-8841	SO-XYZ-20261022-802 / SHP-XYZ-90
GT-XYZ-2026-003	SKU-XYZ-412	PO-XYZ-880355	BATCH-XYZ-BLR-7712	SO-XYZ-20261022-803 / SHP-XYZ-91

DIGITAL CRYPTOGRAPHIC AUDIT SEAL & ENTERPRISE SIGN-OFF

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